

STUDENT ACTIVITY

Advanced Machine Learning Algorithms — Activity

Activity: Algorithm Selection Matrix and Parameter Tuning

Q: Read the three project scenarios below. Based on the specific requirements and strengths of each algorithm, select the most appropriate Machine Learning model (KNN, Naive Bayes, SVM, or Ensemble Techniques) for the task.

Next, in two to three sentences, define what "hyperparameter tuning" is and provide one specific example of a hyperparameter that can be adjusted in the K-Nearest Neighbors (KNN) algorithm.

The Scenarios:

1. **Scenario A:** A technology company needs a fast, probabilistic algorithm to classify millions of user emails into "Spam" or "Not Spam" based on the assumption that all input features (words) are independent of each other.
2. **Scenario B:** A medical research facility needs to classify complex health data into different disease categories by finding the absolute best boundary—or hyperplane—that separates the data classes.
3. **Scenario C:** A financial institution wants to predict loan defaults. Instead of relying on a single model, they want to combine multiple Decision Trees to reduce overfitting, improve prediction stability, and generate the most accurate result possible.

Algorithm Selection Matrix

Scenario	Most Appropriate Algorithm	Reason for Selection
Scenario A: A technology company needs a fast, probabilistic algorithm to classify emails as Spam or Not Spam , assuming all words are independent.	Naive Bayes	Naive Bayes is a probabilistic classifier that assumes features are independent. It is fast, efficient, and widely used for spam email classification .
Scenario B: A medical research facility needs to classify complex health data by finding the best	Support Vector Machine (SVM)	SVM finds the optimal hyperplane that maximizes the margin between different classes, making it highly

Scenario	Most Appropriate Algorithm	Reason for Selection
separating hyperplane between disease categories.		effective for complex classification problems.
Scenario C: A financial institution wants to combine multiple Decision Trees to improve accuracy and reduce overfitting when predicting loan defaults.	Ensemble Techniques (e.g., Random Forest)	Ensemble techniques combine the predictions of multiple Decision Trees to reduce overfitting , improve stability, and achieve higher prediction accuracy.

Hyperparameter Tuning

Hyperparameter tuning is the process of selecting the best values for a machine learning model's parameters before training to improve its performance. These parameters are not learned from the data but are set by the user and can significantly affect the model's accuracy and generalization. In the **K-Nearest Neighbors (KNN)** algorithm, a common hyperparameter is **K**, which specifies the **number of nearest neighbors** used to classify or predict a data point.

Algorithm Selection Matrix

Scenario	Selected Algorithm	Justification
A. Spam Email Classification	Naive Bayes	Uses probability and assumes feature independence, making it ideal for spam detection.
B. Disease Classification	Support Vector Machine (SVM)	Finds the optimal hyperplane to separate different disease classes with high accuracy.
C. Loan Default Prediction	Ensemble Techniques (Random Forest)	Combines multiple Decision Trees to reduce overfitting and improve prediction accuracy and stability.

Hyperparameter Tuning

Hyperparameter tuning is the process of choosing the best parameter values before training a machine learning model to achieve optimal performance. It helps improve the model's accuracy and ability to generalize to new data. In **KNN**, the most common hyperparameter is **K (Number of Nearest Neighbors)**, which determines how many neighboring data points are considered when making a prediction.